

Qualcomm



EdgeFit Coach

Someone is always watching you👁️👁️,

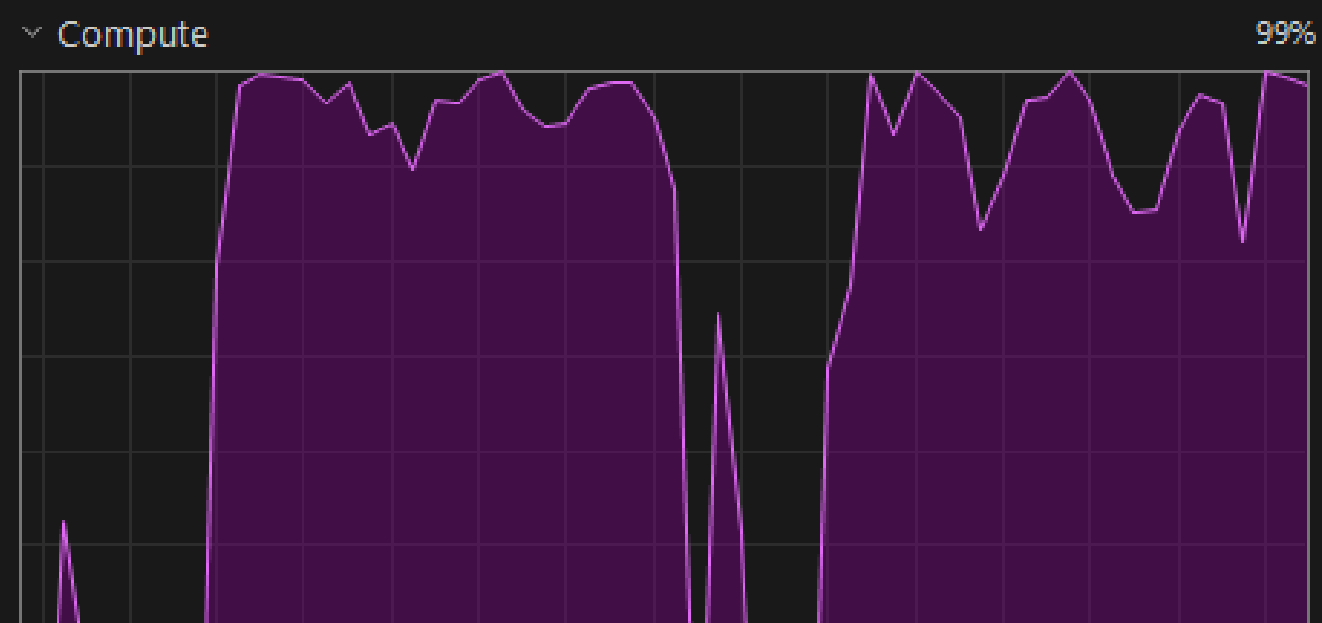
...and this time it's your AI coach making sure your spine lives to see GTA6..

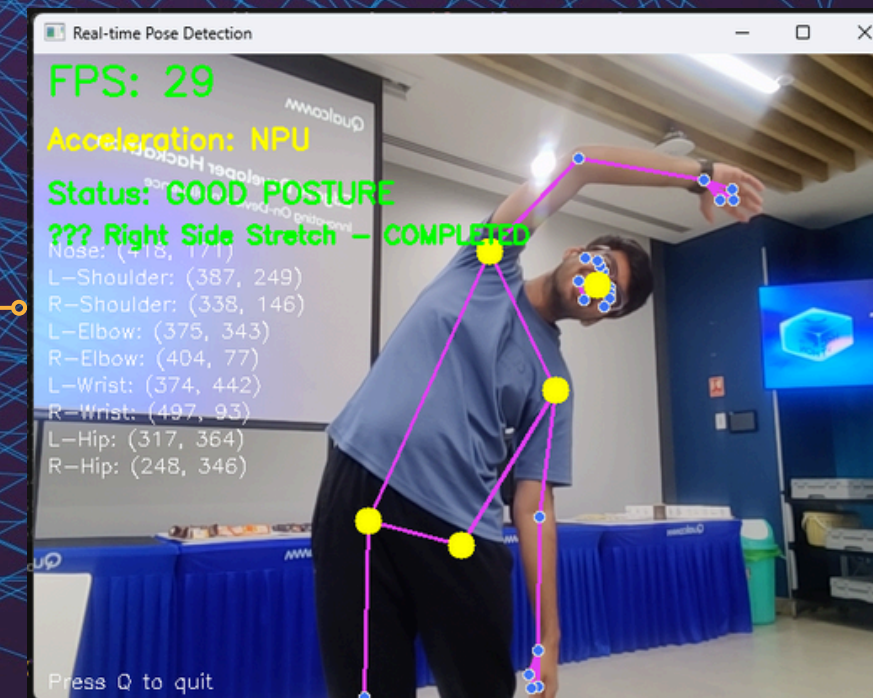
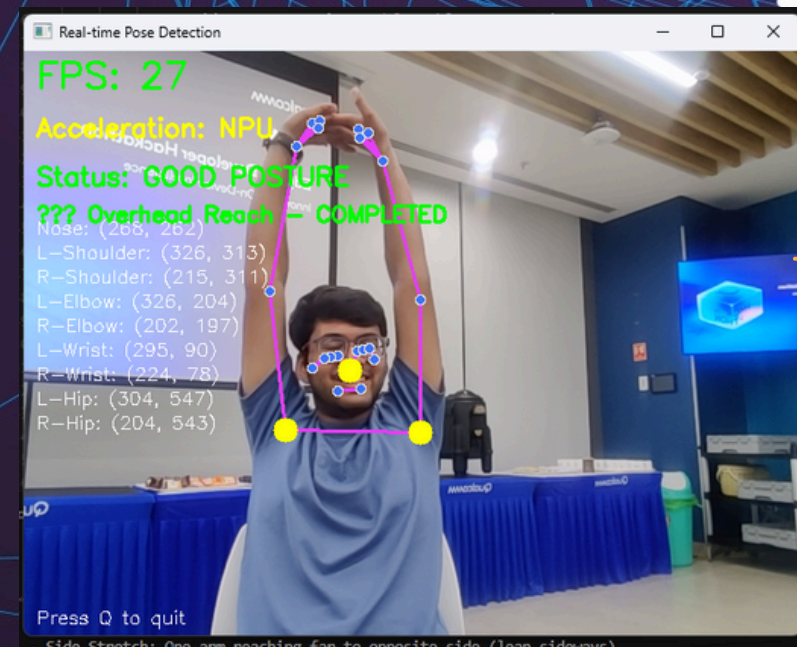
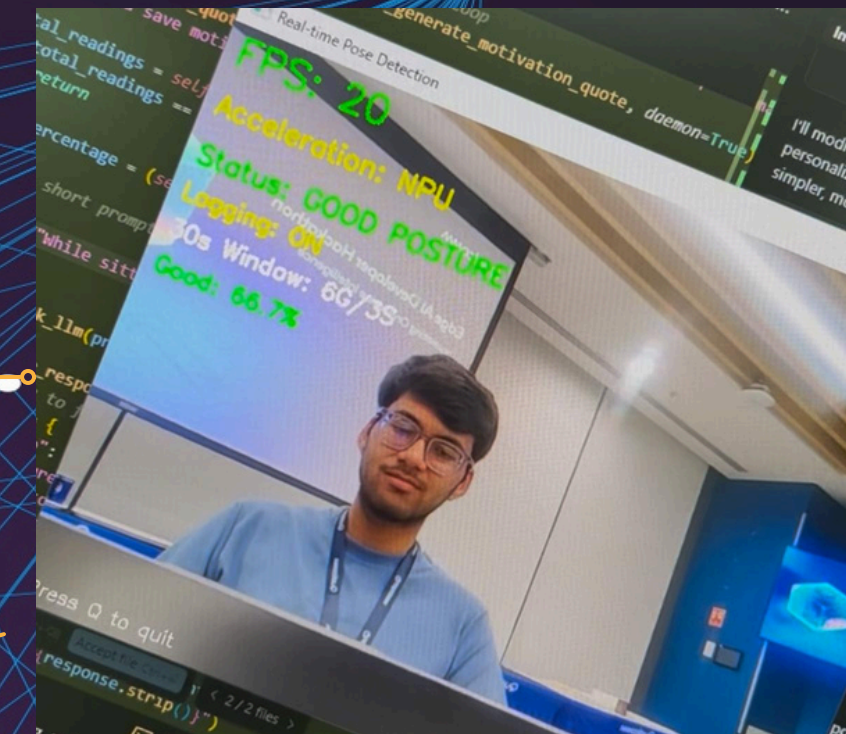
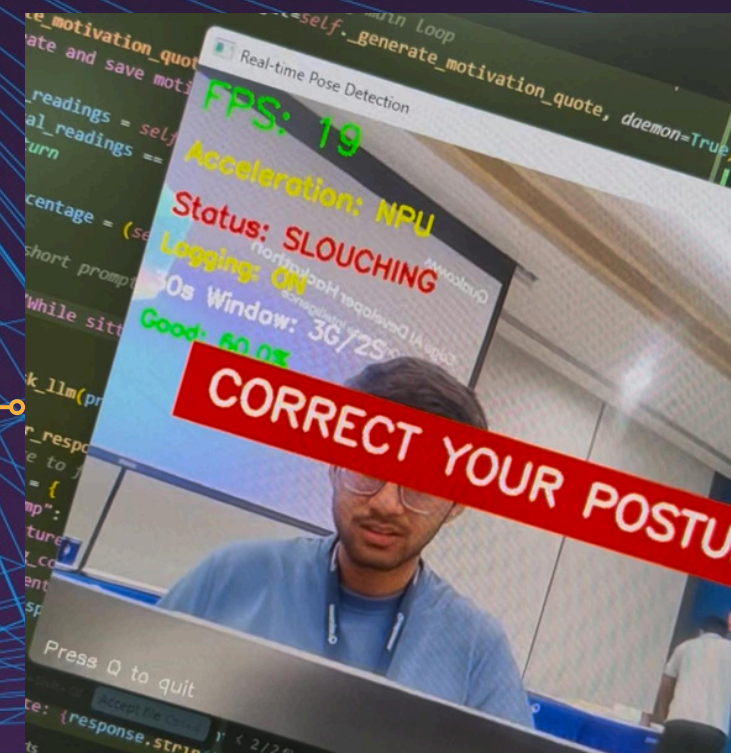
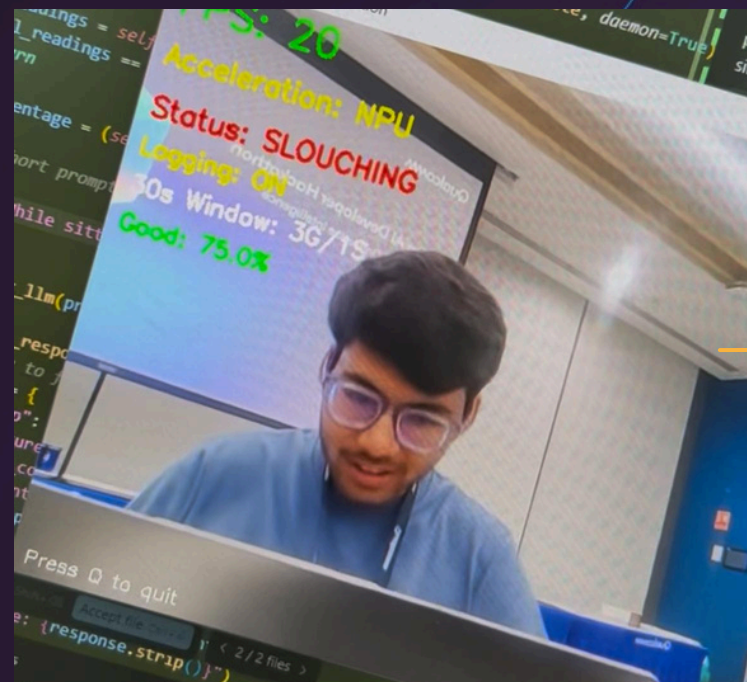
PROBLEM STATEMENT



- Build a full working deployed edge ai app
 - Check✓
- Have something as universal that every person must use as a bare minimum
 - Check✓
- Utilize the on Device NPU for compute
 - Well we fried it✓

NPU Snapdragon(R) X Elite - X1E78100 - Qualcomm(R) Hex...

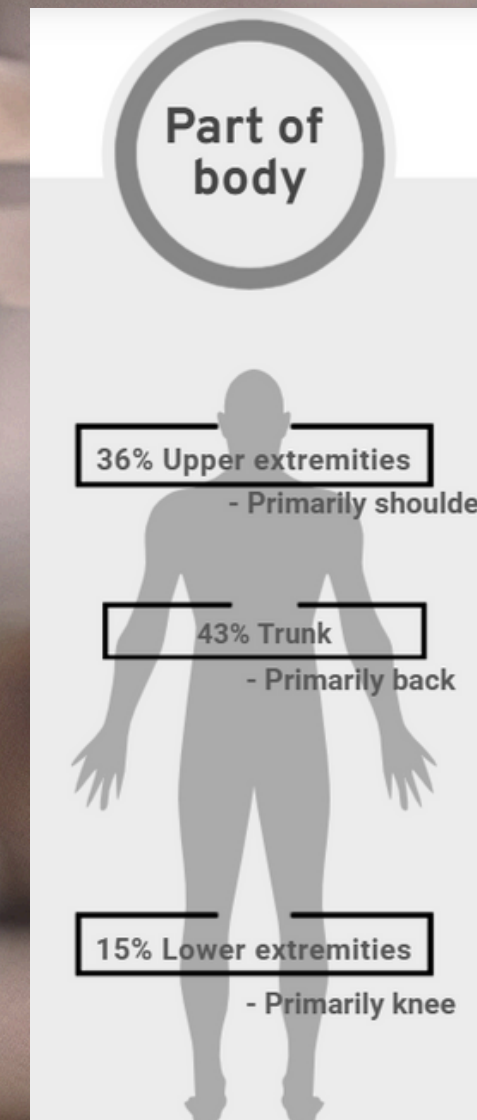
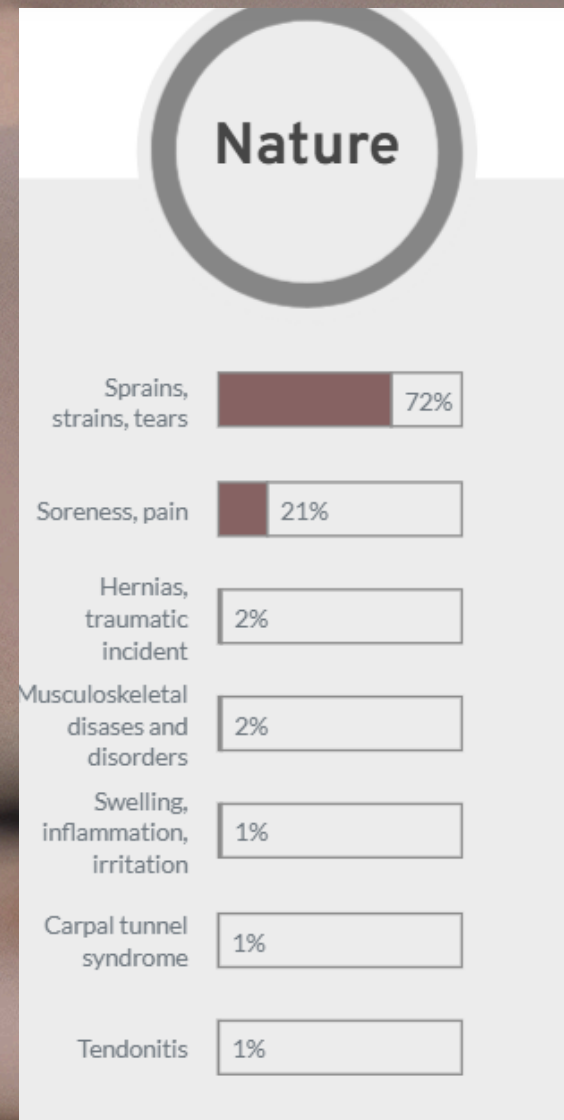




Some real-improvement results –
A transformative journey

Why EdgeFit ?

According to the U.S. Bureau of Labor Statistics (BLS), in the years 2020–2021, the private sector experienced 976,090 musculoskeletal disorder (MSD) cases. Out of all these cases, 72% of all these cases were caused by sprains, strains or tears. Furthermore, 43% of these cases affected the back and 36% affected the neck or the head.



Despite all this paint, the solutions are overpriced chairs – how about an AI that just tells you, 'Sit up!' every 3 minutes



EdgeFit Coach helps you keep track of your daily sitting habits and helps you one step further by motivating you to improve your physical health.

Working hard is important, not having to spend the rest of your life in a bed is even more.

And all of this from the comfort of your device, whatever it may be, even offline. Desktop? Checked. Laptop? Checked. Mobile? No one has good posture while using their mobile.

We won't have your data, but we will have your back.

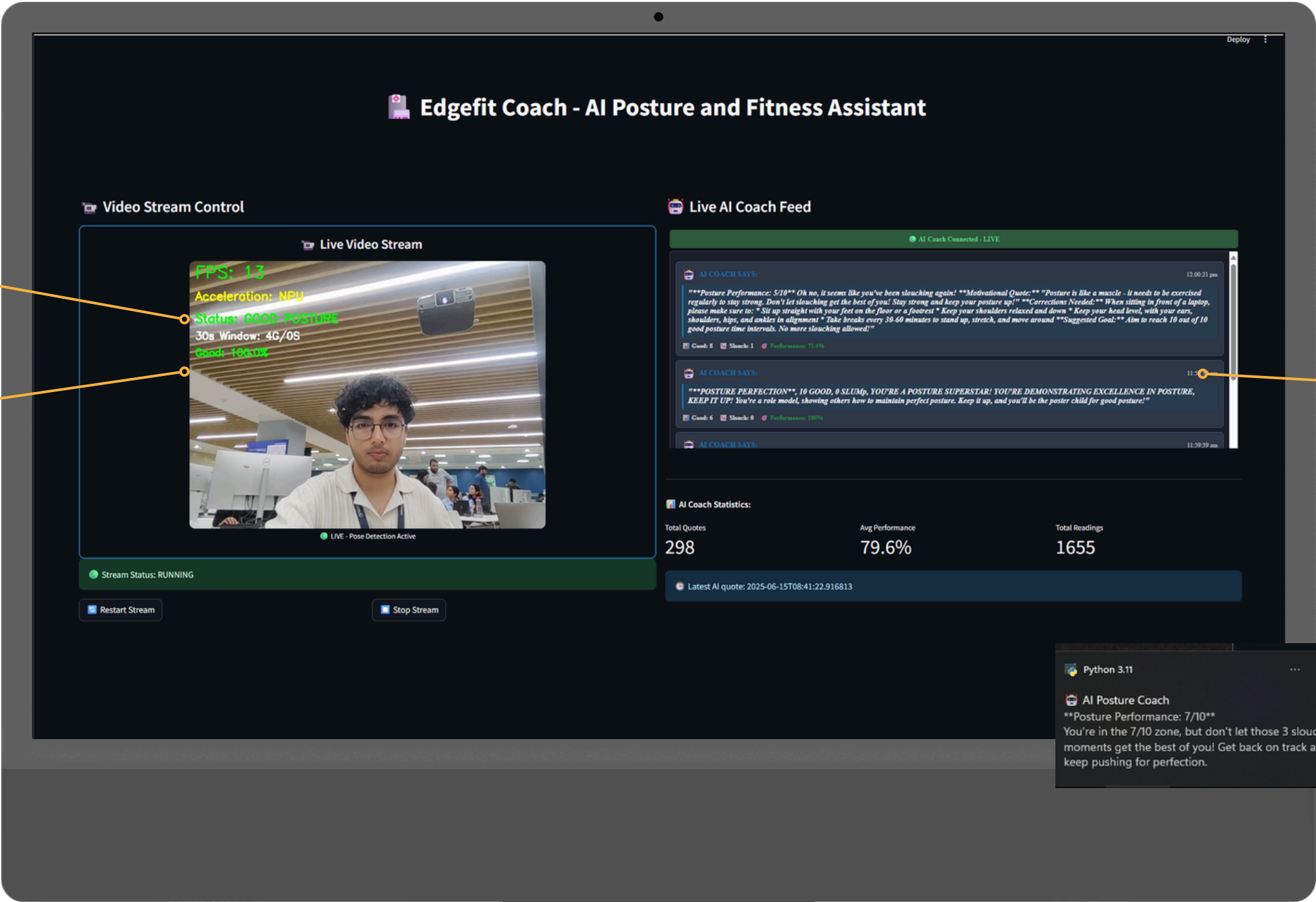
INTERFACE

Live Monitoring

Exercise detected

AI Coach feedback

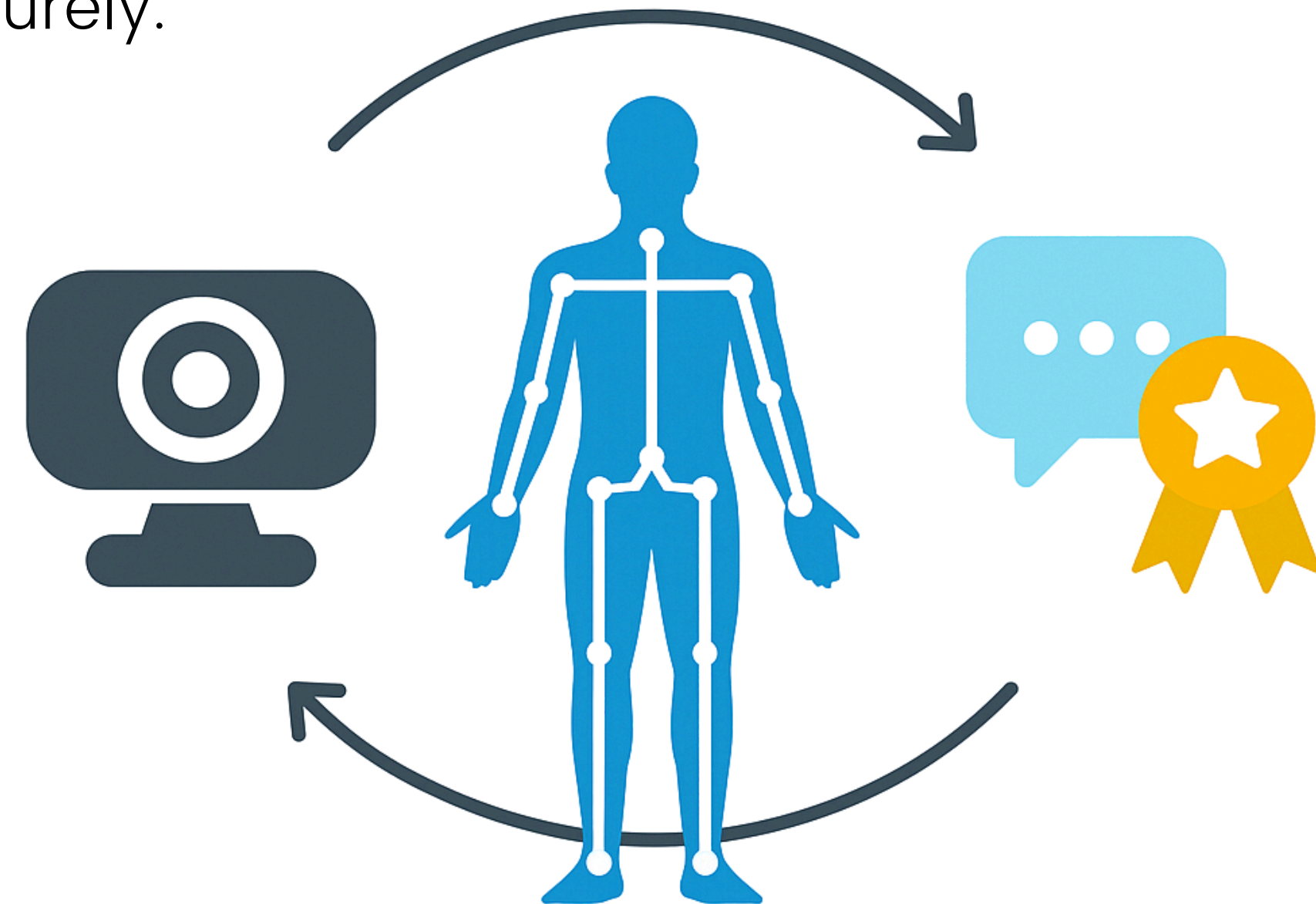
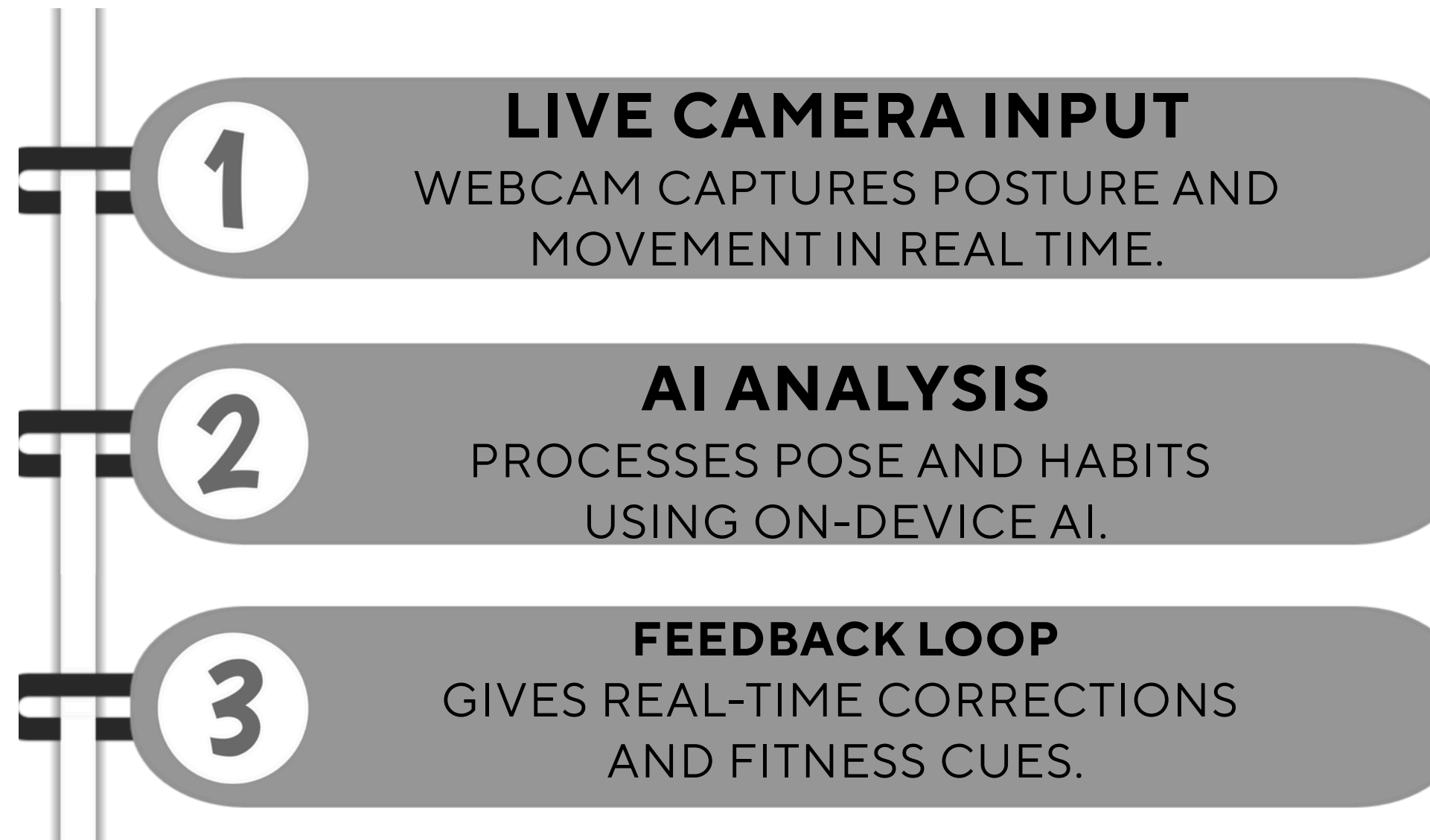
Popup suggestions



OVERVIEW

Now, we explore the core workflow and methodology of EdgeFit Coach. Our system leverages real-time computer vision and on-device AI to improve posture and fitness.

It processes live camera input, analyzes user movement, and delivers instant feedback – privately and securely.



01

Live Pose Monitor

Tracks your posture in real time using the laptop camera. Detects slouching or poor form and provides real-time nudges to correct it, helping you stay aligned throughout the day

02

Posture Correction Popup

LLM reviews your movement patterns to suggest improvements. Whether it's posture, break timing, or exercise quality, feedback feels natural and helpful. Think of it as personalized AI coaching.

03

LLM live suggestions

An on-device LLM provides motivational quotes based on your form. If you're off-track, it nudges you to fix it; if you're aligned, it encourages consistency. Keeps you mentally engaged throughout.

04

Gamified Dashboard

Tracks posture, activity, and workout performance with visual feedback. Points, streaks, and badges make daily movement feel rewarding. Turns fitness into a fun, habit-forming experience.

05

Coach for personalized improvement on analysis

Talk to the LLM for instant posture or fitness guidance. Ask questions, get tips, or request routines—all processed locally for full privacy. Your AI guide is always ready to help.

06

Get dashboard analysis from LLM

You can ask the LLM to explain your dashboard stats. It translates complex posture data into simple insights like “you’ve slouched less today.”

07

Take breaks to exercise with coach

Detects when you’ve been inactive too long and suggests movement. Prompts light stretches or squats to keep your body fresh and focused. A subtle push to stay active during the day.

08

Multilingual chat with coach

Interact with the AI coach in multiple languages for tips, encouragement, or advice — all processed locally to ensure complete privacy.

TECHSTACK

01 MediaPipe-Pose-
Estimation

02 Meta Llama 3.2 3B

03 Sarvam-M

04 Fast API

05 Midas-V2

06 Uvicorn

OUR USP

1

PRIVACY FIRST

NO CLOUD, NO DATA SHARING,

2

SMARTER AI, LESS POWER

ADVANCED MODELS, NPU-OPTIMIZED

3

EXTENDIBLE TO EVERYONE

EVERY IT PROFESSIONAL WILL USE



OUR USES CASES

01 Corporate Posture Monitoring

Used in corporate offices to monitor and correct employee posture during prolonged laptop use. Improves spinal health, reduces fatigue, and promotes productivity—all without sending data to the cloud.

02 Automotive Driver Monitoring

Can be integrated with automotive-grade NPUs like the Qualcomm Snapdragon 8295, as seen in modern vehicles such as the Mahindra BE 6. Detects driver posture, fatigue, or distraction in real time, enhancing road safety while preserving complete data privacy.

03 Hospitals & Rehabilitation Centers

Ideal for physiotherapy and recovery programs. Patients receive instant feedback on their body movement and posture, enabling faster recovery, without the need for constant supervision or network dependency.

04 Selfie Camera Enhancements

Can be used in front-facing cameras to guide users in achieving ideal posture and framing for professional-looking selfies. Runs entirely on-device for real-time feedback with no data sharing.

LIVE DEMO

Edgefit Coach - AI Posture and Fitness Assistant

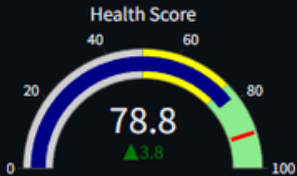
Posture Health Dashboard

Real-time Posture Monitoring Analytics

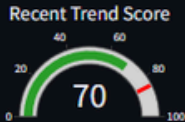
Refresh Data

Video stream paused while viewing dashboard (optimized performance)

Overall Posture Health Score



Performance Metrics



Activity Metrics

Slouch to Good Conversions

Good Posture Time

Conversions

25

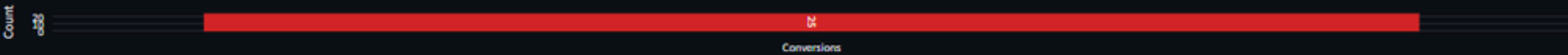
↑ 100.0% of target

Good Posture Minutes

139.8

↑ 100.0% of daily target

Posture Corrections



Minutes Today

139.8 min

Summary

78.8%
Health Score

25
Corrections

73.8%
Consistency

140
Good Minutes



THANK YOU, YOU'RE STILL BEING WATCHED



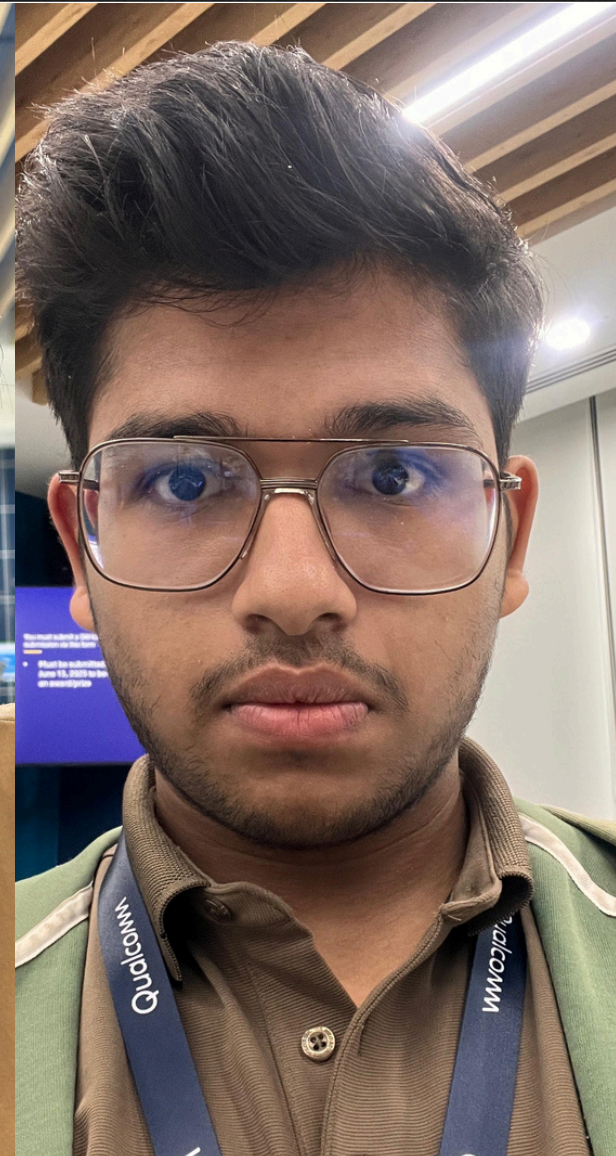
Team Members:

Akshat Gosain

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Parv Jain

Kalash Poddar



EDGEFIT COACH

